

HaulMark Challenge

Dumper Fuel Consumption Prediction Challenge

Mindshift Analytics × AI & SW Guild, IIT Madras

Competition Link: [\[Kaggle Competition\]](#)

Submission Form Link: [\[Submission Link\]](#)

NOTE : Only teams who have submitted the report and codebase will be considered for evaluation.

Submission Deadline: 4th April 2026, End of Day

The Challenge

Given telemetry, route context, and refuelling data, estimate expected fuel consumption and identify abnormal excess fuel use at dumper and route level.

Fuel is one of the **largest operating costs** in open-cast iron ore mining operations. Large fleets of dumpers continuously transport material between excavation faces, stockpiles, crushers, and dump locations, consuming significant volumes of diesel daily.

Improving fuel efficiency directly impacts:

- Operating cost reduction
- Sustainability and emissions

1 Why Fuel Consumption Varies

Fuel consumption in dumpers is influenced by a complex interaction of factors:

1.1 Haul Road Design & Conditions

- Gradient and net lift between loading and dumping points
- Rolling resistance due to road condition
- Curvature and switchbacks
- Haul road maintenance quality

1.2 Operator Behavior

- Acceleration and braking patterns
- Idling and waiting time
- Throttle discipline
- Driving consistency

1.3 Mine Geometry

- Bench structure and elevation differences
- Distance between load and dump points
- Terrain variability

1.4 Operational Dynamics

- Loading efficiency
- Queuing delays
- Cycle time variation

The same dumper operating in the same mine can exhibit significantly different fuel consumption depending on terrain, speed and route taken.

2 Competition Objective

Participants are invited to build models that estimate **fuel consumption of dumpers for haul cycles** between loading and dumping locations using telemetry, spatial data, and operational signals.

Specifically, develop a model that can:

1. Predict **specific fuel consumed** (in litres) for a dumper operating from a load point to a dump point.
2. Predict **generic fuel consumed** (in litres) based on *exposure* (hours of operation), *terrain* (distance, lift, undulations and road zig zag), and *behavior* (speed, stoppage density, etc.).
3. Show match between model predictions and aggregated daily/shift-wise *actual* fuel consumed.

The following parameters are key drivers of fuel consumption:

- Dumper ID and Operator ID
- Engine hours and Idling hours
- Lead distance and Net lift
- Gross elevation change
- Standard deviation of altitude
- Gradient / angle statistics
- Speed statistics
- Stop-go intensity
- Load state (empty or loaded)

3 Data Provided

Participants will be provided with multiple datasets capturing machine behavior, mine layout, and operational events.

3.1 Dumper Telemetry Data

High-frequency telemetry data for each dumper, including:

- Timestamp
- Latitude / Longitude (anonymized)
- Speed, Altitude, Heading / angle
- Ignition status
- Dump signal (in select cases)

This data captures **movement patterns and terrain interaction**.

3.2 Loader / Excavator Telemetry

Telemetry for loading equipment, including position and operating status. This helps in identifying loading locations, inferring loading events, and understanding dumper-loader interaction.

3.3 Refuelling Transactions

Event-level data capturing:

- Timestamp of refuelling
- Dumper ID
- Fuel quantity added

Used to anchor fuel balance, validate consumption estimates, and support aggregation-based evaluation:

$$\text{Actual Fuel} = \text{Opening Level} + \text{Refuelled} - \text{Closing Level}$$

3.4 Dump Switch Analog Signal (Key Input)

Analog dump switch signal is available for some dumpers from **March 1st onward**. This signal captures dumping events via pulse/activation during material discharge.

Why this matters:

- **Accurate Cycle Segmentation:** Each pulse corresponds to a dump event; haul cycles can be reconstructed as: Load → Travel → Dump.

- **Ground Truth for Cycle Boundaries:** Provides explicit end-of-cycle markers, significantly improving segmentation accuracy.
- **Learning Transfer to Historical Data:** Participants can learn cycle patterns from the March 1st onward period and apply this learning to earlier data where the dump switch is unavailable.

3.5 Spatial Data (Shapefiles)

Geospatial layers defining mine structure:

- Loading zones and Dumping zones (OB dump, ROM stock, intermediate stock)
- Haul roads and Benches
- Mine boundary
- Infrastructure (workshop, plant, etc.)

Used for geofencing, route identification, and terrain context.

3.6 Additional Derived Inputs (Participant Driven)

Participants are encouraged to derive:

- Haul cycles (using dump switch and telemetry)
- Distance travelled and net lift
- Elevation variability, stop counts, idle duration
- Speed statistics and route identifiers (cluster-to-cluster)
- Loading and dumping dwell times
- Terrain overlays, haul road snapping, geofence awareness

4 Expected Outputs

4.1 Primary Output (60% Weightage) — Haul Cycle Fuel Prediction

Predict **fuel consumed (litres) for each day**. (Check Kaggle sample submission for detailed submission format.)

Each prediction should include:

- ID
- Predicted fuel consumption on given day (litres)

Develop a **regression model** that estimates fuel consumption aggregated for each day for each dumper, conditioned on exposure, terrain, and operational behavior. The model should generalize across operators, capture route-specific difficulty, and incorporate behavioral variation.

4.2 Secondary Outputs (include in report along with methodology) (20% Weightage)

1. **Route-Level Fuel Benchmark:** Estimate expected fuel consumption for a route independent of dumper.
2. **Dumper Efficiency Component:** Capture dumper-specific or operator-specific variation.
3. **Cycle Segmentation Methodology:** Explain how dump switch data was used and how historical cycles were inferred.
4. **Daily Fuel Consistency:** Ensure aggregated predicted fuel aligns with actual daily fuel consumption per dumper.

4.3 Codebase (20% Weightage)

1. Maintain a clean and readable codebase, preferably on **GitHub**
2. A .py file which can be directly run by us to get your .csv outputs

5 Evaluation

$$\begin{aligned}\text{Score} &= 0.6 \times \text{Raw Prediction Score} \\ &+ 0.2 \times \text{Report and Secondary Outputs} \\ &+ 0.2 \times \text{Codebase and Interview}\end{aligned}$$

- **Raw Prediction Score (60%):** Your predictions will be scored against actual fuel consumption values.
 - Predictions in the format specified by the sample submission file.
 - A `.csv` file with structured prediction outputs.
- **Report (20%):**
 - Clearly explain your pipeline, modeling choices, and feature engineering decisions.
 - Write about your approach towards the secondary outputs and how your model aims to predict those.
- **Codebase & Interview (20%):**
 - Your understanding of the problem and your approach towards solving it will be tested in an interview round during which all team members must be present.
 - Back your decisions with experimental results.
 - Maintain a clean, well-commented codebase.

Top 10 teams in overall score will proceed to Round 2 of the competition!

Note: Qualifying for Round 2 is not a criterion for recruitment to the guild.

6 IP Rights Disclaimer

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By continuing participation, all team members agree to comply with these terms.

7 Frequently Asked Questions

7.1 Data and Access

Q: Can I use external data beyond what is provided?

Answer: For this hackathon, external data is not needed, we have provided all the data.

Q: Is the dump switch signal available for the full dataset?

Answer: No. The dump switch signal is only available from March 1st onward for select dumpers. A key challenge is to transfer cycle-boundary learning to earlier data where this signal is unavailable.

7.2 Evaluation Details

Q: What scoring metric is used for the primary prediction score?

Answer: Root Mean Square Error

Q: Will my code be run during evaluation?

Answer: Yes. Ensure your code is well-organized, reproducible, and includes clear instructions for execution.

7.3 Technical Approach

Q: How should I handle haul cycles for dumpers with no dump switch data?

Answer: This is a core challenge of the competition. You may use spatial geofencing, speed signatures, and patterns learned from dump-switch-equipped dumpers to infer cycle boundaries in historical data.

Q: Should I build one model per route or a single generalized model?

Answer: Both approaches are valid. Experiment with route-conditioned features vs. separate per-route models to determine what generalizes best across your validation set.

Q: How important is feature engineering vs. model selection?

Answer: Both matter, but strong feature engineering—especially around terrain, elevation, and operator behavior—often makes a larger difference in physical-process prediction tasks like this one.

8 Resources and Sample Data

The below directory contains the Sample Dataset and Resources:

[\[folder link\]](#)